

Market Analysis Project Report

A data-driven investigation of order patterns, product performance, and reorder behavior to optimize marketing strategy and improve customer satisfaction

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01-04

Business Overview

Understanding Business Problem, Project Objective & Creating
Analytical Framework

Online grocery platforms generate huge daily logs. Structuring and analysing this raw data transforms it into **valuable business insights** regarding customer behaviour and product performance.

Executive Summary

63,100	49,355	134	21	10.5	59.85%
Customers	Products	Aisles	Departments	Avg Items/Order	Reorder Rate

Three-Stage Integrated Analysis

This report presents a comprehensive analysis of online grocery ordering behavior covering **63,100 customers**, **49,355 products**, **134 aisles**, and **21 departments**.

The analysis was conducted across three integrated stages:

- 1. Data validation and exploration in **MySQL**
- 2. Statistical and behavioral analysis in **Python**
- 3. Executive visual reporting in **Power BI**

Key Findings at a Glance

The business is a **high-frequency, low-basket-value** grocery operation built around fresh, perishable staples.

- **Produce and dairy** dominate order volume and reorder rate
- Peak ordering: **9 AM - 5 PM**, highest **Sun-Mon**
- Perishable categories show **3-4x higher** reorder rates
- A small set of SKUs carries disproportionate demand
- Clear time-of-day and day-of-week seasonality

Single Most Decision-Ready Finding: Convergent evidence across four independent metrics confirms that a small set of perishable staples anchors customer engagement, while the majority of the 49,355-product catalogue sees minimal repeat engagement.

Project Overview & Background

Online grocery platforms generate large volumes of transactional data daily — what was ordered, when, by whom, how it was categorized, and whether it was purchased again.

On its own, this data is simply a log of activity. Its business value only emerges when it is **validated, structured, and analyzed** to reveal repeatable patterns in customer behavior and product performance.

Dataset Structure

Relational dataset (**project_orders**) modelled on the Instacart Market Basket dataset, comprising **five linked tables**:

aisles | departments | products | orders | order_products_train

The order_products_train table maps products to orders flagged with **eval_set = "train"** — the labelled, analysis-ready slice.

Three-Stage Engagement Structure

01 MySQL Validation

Connect to hosted database, validate data integrity, perform first-pass exploration. Check schema, row counts, relationships.

02 Python Analysis

Merge five tables into analytical dataset. Answer 20 business questions with statistical analysis and exploratory charts.

03 Power BI Dashboards

Convert processed output into four interactive executive dashboards for non-technical stakeholders and decision-makers.

Analytical Discipline: Every finding was pushed one step further than "what happened" to explain **why it matters** and **what action it implies**. Each of the 20 brief questions was addressed using the same three-layer approach: **SQL for validation, Python for quantitative analysis, and Power BI for stakeholder-facing visualization**.

Business Problem

Grocery e-commerce operates on thin margins and high order frequency, which means small improvements in basket size, reorder rate, or delivery-slot efficiency compound into meaningful revenue impact. Without a structured view of purchasing behavior, Marketing and Merchandising teams are left to make decisions based on intuition rather than evidence.

CORE PROBLEM

How can historical order data be used to optimize marketing strategy and improve customer satisfaction?

Three Linked Analytical Questions

What happened?

What are the dominant patterns in product demand, reorder behavior, and order timing?

Why did it happen?

What structural or behavioral factors explain those patterns?

What should stakeholders do?

Which concrete, data-backed actions should Marketing, Merchandising, and Operations take?

Scope of Decisions Addressed: Which products to promote, which time slots to prioritize, which customers are at risk of churn, how to allocate staffing and delivery resources, where to rationalize the product catalogue, and how to design replenishment-based marketing programs (e.g., subscribe-and-save on top reorder items).

Project Objectives & Analytical Framework

The project brief specified **20 analytical questions** spanning product catalogue structure, customer behavior, order timing, and reorder patterns. These were grouped into **four analytical objectives** that map directly to business decisions:

Catalogue & Assortment Objective

Questions 1-3, 20

Quantify product distribution across departments and aisles. Identify which categories carry the most SKUs versus the most demand.

Customer Behavior Objective

Questions 5, 6, 10, 11, 14, 19

Profile customer base by order frequency, basket size, department reach, and purchase cadence.

Temporal / Demand-Timing Objective

Questions 7, 8, 15, 16, 18

Identify peak ordering hours and days, and how basket size shifts across the week, to inform staffing and promotional scheduling.

Product Performance & Loyalty Objective

Questions 4, 9, 12, 13, 17

Surface products and aisles with highest reorder rates, and categories that anchor repeat purchase behavior.

Unified Analytical Discipline

Each objective is addressed using the same three-layer discipline: **SQL for validation**, **Python for quantitative analysis**, and **Power BI for stakeholder-facing visualization** — with every finding pushed one step further than "what happened" to explain why it matters and what action it implies. This ensures triangulated, cross-checked conclusions that can be trusted for business decision-making.

Dataset Overview — Five-Table Relational Schema

Table	Description	Records	Primary Key
Aisles	Aisle reference list (aisle_id -> aisle name)	134	aisle_id
Departments	Department reference list (dept_id -> dept name)	21	department_id
Products	Product catalogue (product_id, name, aisle_id, dept_id)	49,355	product_id
Orders	One row per order (user_id, eval_set, order timing)	1,048,575	order_id
Order Products (Train)	One row per product within an order (order_id, product_id, cart, reordered)	1,048,575	order_id + product_id

Key Structural Points

Train-set scope: The order_products table is explicitly named order_products_train and only maps to orders where eval_set = "train". This is a deliberate, labelled analytical subset (consistent with the standard Instacart dataset design), not the full order history.

After merging, the analytical base table (master_df) contains **318,051 order-product line items**.

Row-count ceiling: Both the orders and order_products_train tables report exactly 1,048,575 rows — one short of 2^20 (1,048,576), the well-known row-import ceiling of tools such as Excel/CSV-based bulk loaders.

This suggests the source tables were **truncated at import** rather than reflecting a naturally occurring row count. Noted as a data-limitation caveat.

Table Relationships: **order_id** connects Orders and Order Products | **product_id** connects Order Products and Products | **aisle_id** connects Products and Aisles | **department_id** connects Products and Departments. These relationships allow comprehensive analysis combining customer, product, and order information.

Dataset Schema & Data Quality Results

Data Quality Result

Independent validation in both SQL and Python confirmed **zero missing values** and **zero duplicate primary keys** across all five tables. Referential integrity held across every join. One exception: an aisle labelled "**missing**" (**aisle_id 100**) containing 1,254 products — a real category value, not a null.

Entity Relationship Overview

Relationship	Join Key	Cardinality
Orders ↔ Order Products	order_id	1 : Many
Order Products ↔ Products	product_id	Many : 1
Products ↔ Aisles	aisle_id	Many : 1
Products ↔ Departments	department_id	Many : 1

Validation Checks Performed

- ✓ Primary key uniqueness verified (product_id, order_id, aisle_id, dept_id)
- ✓ Foreign key referential integrity confirmed across all joins
- ✓ NULL value scan across all columns in all five tables
- ✓ Duplicate record audit using primary key constraints
- ✓ Complete database join combining all five tables validated
- ✓ Row counts independently confirmed in SQL and Python

"Missing" Aisle Note: During validation, aisle_id 100 with the name "missing" was identified containing 1,254 products. This was verified as an **actual aisle label** in the source data rather than a data quality issue. The products in this aisle were treated as a legitimate category throughout the analysis. All findings in this report describe patterns within the train-set slice (master_df: 318,051 rows x 15 columns).

Tools & Technology Stack

Layer	Tool / Technology	Purpose in this Project	Key Functions
Database	MySQL (hosted, project_orders)	Source of truth; data validation and first-pass exploration	Schema inspection, row counts, integrity checks, JOIN validation
Analysis	Python (Google Colab): Pandas, NumPy	Table merging, aggregation, and the 20-question analysis	Data merging, groupby aggregation, statistical analysis
Visualization (Analysis)	Matplotlib	Exploratory charts for peak hours and day-of-week analysis	Bar charts, order distribution plots
Visualization (Reporting)	Microsoft Power BI	Interactive executive dashboards for non-technical stakeholders	KPI cards, filters, cross-visual interactions
Data Interchange	CSV (UTF-8)	Bridge format between Python analysis and Power BI	Cleaned tables, derived results, master dataset export

Technical Environment Details

Python Environment: Google Colab with standard data science stack

- **Pandas** - Data manipulation and aggregation
- **NumPy** - Numerical computations
- **SQLAlchemy / PyMySQL** - Database connectivity
- **Matplotlib** - Data visualization

Database Environment: Hosted MySQL instance (project_orders)

- Five interconnected relational tables
- Primary and foreign key constraints
- Direct SQLAlchemy connection from Python
- Eval_set filtering for train-set isolation

Note: The CSV export layer ensures reproducibility and enables the Power BI dashboards to be refreshed as new data becomes available.

Methodology — Six-Stage Analytics Pipeline

- 1

Connect & Explore (SQL)
Connect to hosted MySQL. Inspect schema (SHOW TABLES, DESCRIBE).
Capture row counts for all five tables.
- 2

Validate (SQL)
Check every table for missing values, duplicate primary keys, and referential integrity via join tests.
- 3

Load & Merge (Python)
Connect via SQLAlchemy/PyMySQL. Filter eval_set="train". Merge into master_df (318,051 rows x 15 cols).
- 4

Analyze (Python)
Answer all 20 brief questions using groupby/aggregation. Produce summary statistics and exploratory charts.
- 5

Export (Python -> CSV)
Export cleaned tables and key results to CSV as interchange layer into the visualization tool.
- 6

Visualize (Power BI)
Build four dashboards: Executive, Customer, Product Performance, and Department & Category Analysis.

Pipeline Outputs

SQL Validation Output:

Validated dataset integrity across 5 tables. Zero missing values, zero duplicates, full referential integrity confirmed.

Python Analysis Output:

master_df (318,051 rows x 15 columns). Answers to all 20 analytical questions with statistical evidence. Two exploratory charts (peak hours, day-of-week volume).

CSV Export Output:

Aisles, Departments, Products, Orders, Order Products, Master Dataset, Top Reordered Products, Most Reordered by Department.

Power BI Output:

4 interactive dashboards with calculated measures, KPI cards, filters, and cross-visual interactions.

Business Deliverables:

Comprehensive insights, 12 recommendations, and expected impact assessment for Marketing, Merchandising, and Operations.

05

SQL Validation

Database Exploration, Data Quality Checks & Relationship
Validation

Before any behavioral analysis, the raw dataset was validated directly in MySQL. This stage answers a single question that every later insight depends on: **can this data be trusted?**

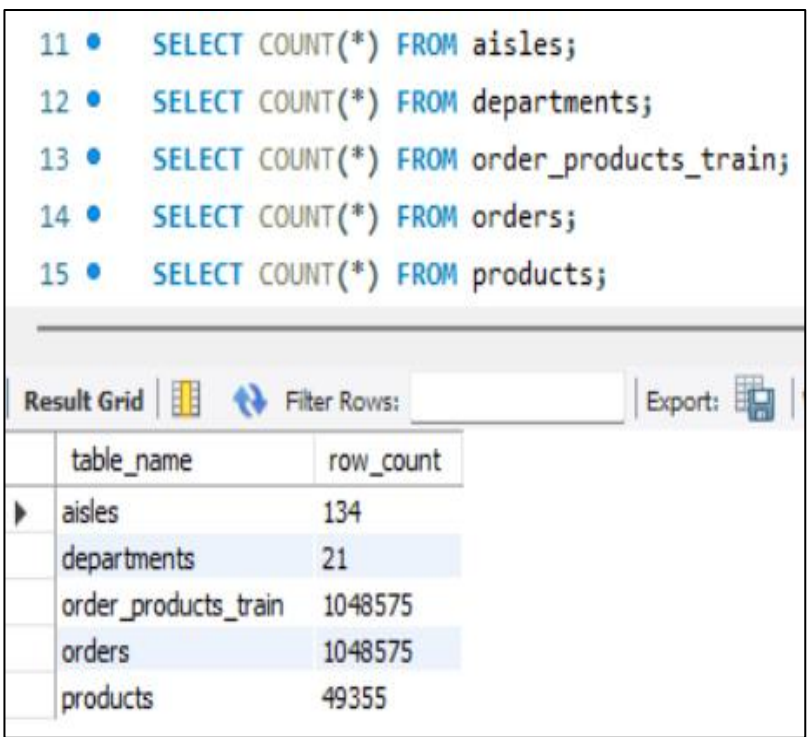
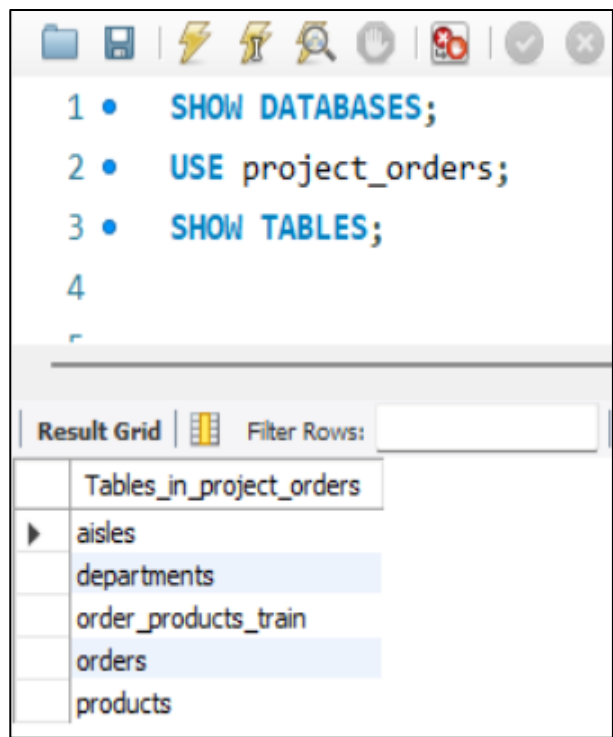
SQL Validation — Database Connection & Exploration

Database Connection

Connected to the MySQL database provided for the project. The database contained five interconnected tables. To confirm a successful connection and verify available schemas, the following commands were executed:

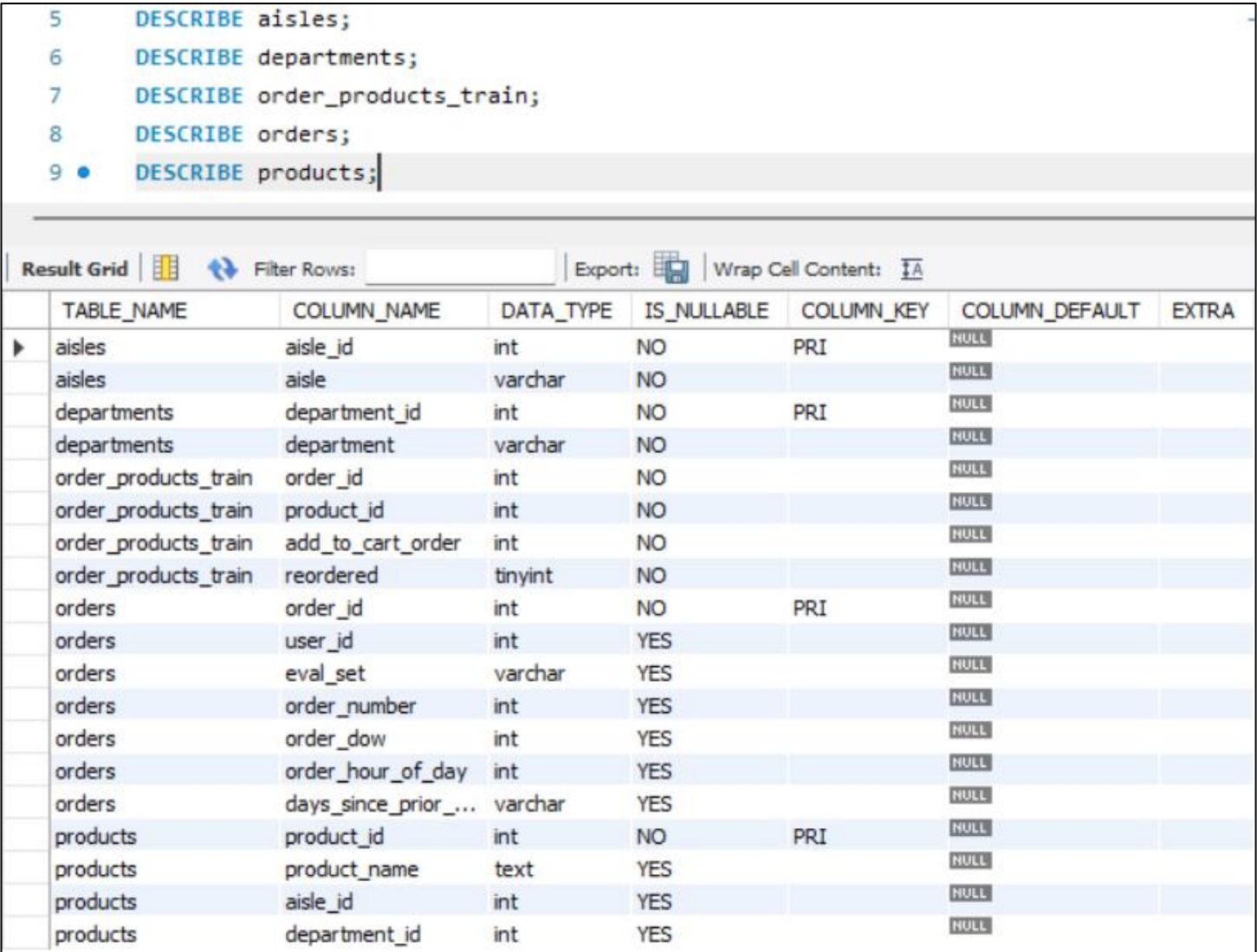
SHOW DATABASES; | **USE project_orders;**

SQL EXPLORATION SCREENSHOTS



Database Exploration

Schema explored using **SHOW TABLES** and **DESCRIBE** to identify structure, primary keys, and data types. Row counts obtained for all tables. Sample records inspected to verify data completeness and correct structure.



SQL Validation — Sample Data Verification

Sample records from each table were retrieved using `SELECT * FROM <table> LIMIT 5` to verify structure, data format, and relationships. The following tables were explored:

Aisles: Contains unique aisle identifiers and corresponding aisle names. **Departments:** Contains department identifiers and names. **Products:** Stores product information with associated aisle and department IDs. **Orders:** Contains customer order details (user ID, order sequence, day, hour, days since prior). **Order_Products_Train:** Order-product relationships with cart sequence and reordered flag.

SAMPLE RECORDS FROM FIVE TABLES

20 • SELECT * FROM orders LIMIT 5;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell

	order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order
▶	6	22352	prior	4	1	12	30
	8	3107	prior	5	4	6	17
	13	45082	prior	2	6	17	1
	14	18194	prior	49	3	15	3
	15	54901	prior	51	3	11	2
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exploration Summary

Table	Row Count	Primary Key	Validation Status
aisles	134	aisle_id	Unique, no nulls
departments	21	department_id	Unique, no nulls
products	49,355	product_id	Unique, no nulls
orders	1,048,575	order_id	Unique, no nulls

Verification Result: The sample records confirmed that the database structure matched the provided schema and that all tables contained valid and consistent data. Primary keys were unique, foreign keys aligned correctly, and data types were appropriate for analysis. This verification ensured the dataset was ready for table joins and further analytical processing.

SQL Validation — Missing Value Analysis

✓ **Result: No Missing Values Found.** Every column across all five tables was scanned using targeted aggregation queries. No missing values were found in the dataset.

Process:

- Checked all columns across all five tables for NULL values
- Used targeted aggregation queries (COUNT(*) vs COUNT(column))
- Verified in both SQL and Python independently
- No additional data cleaning required

Implication:

The dataset is clean and analysis-ready. Since no missing values or duplicate records were found, no additional data cleaning was required before proceeding with SQL and Python analysis. This validates the dataset's reliability for business decision-making.

SQL NULL CHECK QUERIES & RESULTS

```
23 • SELECT
24   SUM(aisle_id IS NULL) AS aisle_id_null,
25   SUM(aisle IS NULL) AS aisle_null
26 FROM aisles;
27 • SELECT
28   SUM(department_id IS NULL) AS department_id_null,
29   SUM(department IS NULL) AS department_null
30 FROM departments;
31 • SELECT
32   SUM(product_id IS NULL) AS product_id_null,
33   SUM(product_name IS NULL) AS product_name_null,
34   SUM(aisle_id IS NULL) AS aisle_id_null,
35   SUM(department_id IS NULL) AS department_id_null
36 FROM products;
```

```
37 • SELECT
38   SUM(order_id IS NULL) AS order_id_null,
39   SUM(user_id IS NULL) AS user_id_null,
40   SUM(eval_set IS NULL) AS eval_set_null,
41   SUM(order_number IS NULL) AS order_number_null,
42   SUM(order_dow IS NULL) AS order_dow_null,
43   SUM(order_hour_of_day IS NULL) AS order_hour_null,
44   SUM(days_since_prior_order IS NULL) AS days_null
45 FROM orders;
46 • SELECT
47   SUM(order_id IS NULL) AS order_id_null,
48   SUM(product_id IS NULL) AS product_id_null,
49   SUM(add_to_cart_order IS NULL) AS cart_order_null,
50   SUM(reordered IS NULL) AS reordered_null
51 FROM order_products_train;
```

Result Grid				Filter Rows:	Export:
	table_name	column_name	null_count		
▶	aisles	aisle_id	0		
	aisles	aisle	0		
	departments	department_id	0		
	departments	department	0		
	products	product_id	0		
	products	product_name	0		
	products	aisle_id	0		
	products	department_id	0		
	orders	order_id	0		
	orders	user_id	0		
	orders	eval_set	0		
	orders	order_number	0		
	orders	order_dow	0		
	orders	order_hour_o...	0		
	orders	days_since_p...	0		
	order_products_train	order_id	0		
	order_products_train	product_id	0		
	order_products_train	add_to_cart_...	0		
	order_products_train	reordered	0		

SQL Validation — Duplicate Record Verification

✓ **Result: No Duplicate Records.** Verified duplicate records using Primary Keys and appropriate key combinations. No duplicate records were identified across any table.

Process:

- Audit conducted on all five tables
- Primary key constraints verified (no overlaps)
- Composite keys checked where applicable
- Prevents skewed aggregations and duplicate counting

Implication:

All validation checks confirmed the dataset is clean, consistent, and suitable for further analysis. The absence of duplicates ensures that aggregation queries produce accurate counts and that business insights are not distorted by redundant data.

SQL DUPLICATE CHECK QUERIES & RESULTS

53 •

SELECT COUNT(*) - COUNT(DISTINCT aisle_id) AS duplicate_aisle_id

54

FROM aisles;

55 •

SELECT COUNT(*) - COUNT(DISTINCT department_id) AS duplicate_department_id

56

FROM departments;

57 •

SELECT COUNT(*) - COUNT(DISTINCT product_id) AS duplicate_product_id

58

FROM products;

59 •

SELECT COUNT(*) - COUNT(DISTINCT order_id) AS duplicate_order_id

60

FROM orders;

61 •

SELECT COUNT(*) AS total_rows,

62

COUNT(DISTINCT order_id, product_id) AS unique_order_product_pairs

63

FROM order_products_train;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	metric	count_value
▶	duplicate_aisle_id	0
	duplicate_department_id	0
	duplicate_product_id	0
	duplicate_order_id	0
	total_rows	1048575
	unique_order_product_pairs	1048575

SQL Validation — Table Relationship Validation

SQL JOIN operations were performed to validate relationships among the five tables. Products were linked with aisles and departments; customer orders were connected with purchased products. These joins confirmed database relationships were accurate.

Validation Summary:

- ✓ All foreign key relationships validated successfully
- ✓ No orphaned records or broken links found
- ✓ Perfect referential integrity for deep-dive analysis
- ✓ Complete database join produces unified analytical view
- ✓ Dataset confirmed clean, consistent, analysis-ready

Relationship	Join Key	Status
Products ↔ Aisles	aisle_id	Verified
Products ↔ Departments	department_id	Verified
Orders ↔ Order_Products_Train	order_id	Verified
Complete 5-Table Join	All keys	Verified

SAMPLE OF SQL JOIN VALIDATION SCREENSHOTS

77 •

SELECT

o.order_id, o.user_id,

p.product_name, a.aisle,

d.department, op.reordered

FROM order_products_train op

JOIN products p ON op.product_id = p.product_id

JOIN aisles a ON p.aisle_id = a.aisle_id

JOIN departments d ON p.department_id = d.department_id

JOIN orders o ON op.order_id = o.order_id LIMIT 10;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	order_id	user_id	product_name	aisle	department	reordered
▶	38	42756	Shelled Pistachios	nuts seeds dried fruit	snacks	0
	38	42756	Organic Biologique Limes	packaged vegetables fruits	produce	0
	38	42756	Organic Raw Unfiltered Apple Cider Vinegar	oils vinegars	pantry	0
	38	42756	Organic Baby Arugula	packaged vegetables fruits	produce	1
	38	42756	Organic Hot House Tomato	fresh vegetables	produce	0
	38	42756	Green Peas	frozen produce	frozen	0
	38	42756	Bunched Cilantro	fresh herbs	produce	0
	38	42756	Flat Parsley, Bunch	fresh herbs	produce	0
	38	42756	Fresh Dill	fresh herbs	produce	0
	96	17227	Roasted Turkey	lunch meat	deli	1

06-08

Python Data Analysis

Data Preparation, 20 Analytical Queries, Interpretations &
Business Insights

After SQL validation, analysis shifted to **Python** for advanced data manipulation. All **20 business questions** were answered with supporting code, statistical analysis, and actionable business insights.

Python Data Preparation

After SQL validation, analysis shifted to **Python (Google Colab)** for efficient handling of large datasets and advanced data manipulation. Python provides greater flexibility for processing, aggregation, and analysis compared to SQL.

Preprocessing Steps Performed

1. Installed required Python libraries
2. Imported Pandas, NumPy, SQLAlchemy
3. Established MySQL database connection
4. Imported all five tables into DataFrames
5. Verified dimensions of each table
6. Verified column names and data types
7. Checked for missing values (independent verification)
8. Checked for duplicate records
9. Merged five tables into single master dataset
10. Used primary/foreign keys for joins
11. Confirmed merged dataset suitability
12. Filtered to eval_set = "train"

SAMPLE OF OPERATION LIKE MERGE

```
orders_train = orders[orders["eval_set"] == "train"].copy()
master_df = (
    order_products
    .merge(orders_train, on="order_id", how="inner")
    .merge(products, on="product_id", how="left")
    .merge(aisles, on="aisle_id", how="left")
    .merge(departments, on="department_id", how="left"))
```

Result: The merged dataset (master_df) contains **318,051 order-product line items** across **15 columns**, combining customer, product, aisle, department, and order timing data into a unified analytical base table ready for all 20 business questions.

Query 1 — Top 10 Aisles with Highest Number of Products

Objective: Identify aisles with most products | Catalogue & Assortment Objective

PYTHON CODE

```
q1_top_aisles = (  
    products.merge(aisles, on="aisle_id")  
    .groupby("aisle", as_index=False)  
    .agg(total_products=("product_id","count"))  
    .sort_values("total_products", ascending=False))  
  
# Verify what "missing" actually means  
aisles[aisles["aisle"] == "missing"]
```

Note: "missing" (aisle_id=100) is an actual aisle name, not incomplete data. Verified separately.

Interpretation

The "missing" aisle contains the most products (1,254). Candy Chocolate, Ice Cream, Vitamins & Supplements, and Yogurt also have large variety. Distribution indicates certain aisles have significantly broader assortment.

#	Aisle	Total Products
1	missing	1,254
2	candy chocolate	1,235
3	ice cream ice	1,084
4	vitamins supplements	1,030
5	yogurt	1,023
6	chips pretzels	980
7	packaged cheese	888
8	tea	887
9	frozen meals	880
10	cookies cakes	869

Business Insight: Aisles with larger product assortment require improved inventory planning, shelf management, and category organization. High-product aisles should receive greater attention during inventory replenishment and promotional planning.

Query 2 — Total Number of Unique Departments

Objective: Determine total unique departments | Catalogue & Assortment Objective

PYTHON CODE

```
q2_department_count = departments["department"].nunique()
print(f"Total Unique Departments: {q2_department_count}")
```

Output: Total Unique Departments: 21

21

Unique Departments

Interpretation

The dataset contains **21 unique departments**, representing major product categories. Each department groups products with similar characteristics, making inventory organization and performance analysis easier. This confirms the store offers a wide variety of categories.

All 21 Departments: produce, dairy eggs, snacks, beverages, frozen, pantry, household, bakery, canned goods, dry goods pasta, deli, personal care, meat seafood, breakfast, international, missing, babies, alcohol, pets, other, bulk

Business Insight: The presence of 21 departments indicates a diverse product portfolio. Department-level analysis can help management identify high-performing and low-performing categories, optimize inventory allocation, and design targeted marketing strategies for different customer segments.

Query 3 — Distribution of Products Across Departments

Objective: Analyze product distribution across departments | Catalogue & Assortment Objective

PYTHON CODE

```
q3 = (products.merge(departments, on="department_id")
      .groupby("department", as_index=False)
      .agg(total_products=("product_id", "count"))
      .sort_values("total_products", ascending=False))
```

#	Department	Total Products
1	personal care	6,524
2	snacks	6,221
3	pantry	5,331
4	beverages	4,324
5	frozen	3,982
6	dairy eggs	3,428
7	household	3,068
8	canned goods	2,073
9	dry goods pasta	1,845
10	Produce	1,671
...	(11 more)	...

Interpretation

Personal Care has the largest assortment (6,524), followed by Snacks (6,221) and Pantry (5,331). Produce, Dry Goods, and Canned Goods contain fewer products. Distribution is highly uneven — certain departments offer much broader variety.

Business Insight: The product catalogue is concentrated in a few major departments. Personal Care, Snacks, and Pantry require greater attention in inventory planning, category management, and shelf space allocation due to their extensive product range.

Query 4 — Top 10 Products with Highest Reorder Rates

Objective: Identify highest reorder-rate products (min 100 orders) | Product Performance & Loyalty

PYTHON CODE

```
def top_reorder_products(min_orders):
    result = master_df.groupby("product_name")
    .agg(total_orders=("product_id","count"),
         reorder_count=("reordered","sum"))
    result["reorder_rate"] = (result["reorder_count"]
                             / result["total_orders"] * 100).round(2)
    return result[result.total_orders >= min_orders]
    .sort_values(["reorder_rate","total_orders"],
                 ascending=[False,False]).head(10)
```

Threshold = 100 orders: Filters out statistically weak results from small samples, ensuring reliable business decisions.

Interpretation

Organic milk products consistently rank highest across all thresholds. Banana remains in top 10 despite thousands of orders, confirming broad-based popularity. Threshold of 100 balances sample size with reliability.

#	Product Name	Orders	Reorders	Rate
1	Organic Low Fat Milk	100	93	93.00%
2	Organic Whole Milk DHA	115	104	90.43%
3	Milk Organic Vitamin D	139	124	89.21%
4	Organic Reduced Fat Milk	274	242	88.32%
5	Banana	4,357	3,840	88.13%
6	Pure Sparkling Water	258	222	86.05%
7	Bag of Organic Bananas	3,522	3,016	85.63%
8	Organic Whole Milk	1,121	957	85.37%
9	Organic Fat Free Milk	188	160	85.11%
10	Organic Large Brown Eggs	257	216	84.05%

Business Insight: Organic dairy products, bananas, and sparkling water are highly loyal products that should be prioritized for inventory planning, promotional campaigns, and customer retention strategies. A minimum order threshold of 100 was selected for final analysis.

Query 5 — Total Number of Unique Users

Objective: Determine total unique customers | Customer Behavior Objective

PYTHON CODE

```
q5_unique_users = orders["user_id"].nunique()
print(f"Total Unique Users: {q5_unique_users}")
```

Output: Total Unique Users: 63100

63,100

Unique Customers

Interpretation

The dataset contains **63,100 unique customers** who have placed orders. This indicates a large and diverse customer base, making the dataset suitable for customer behavior and purchasing pattern analysis. A larger customer base improves the reliability of analytical insights.

Business Insight: The presence of over 63,000 unique customers indicates a broad market reach. This provides opportunities for customer segmentation, personalized marketing campaigns, loyalty programs, and targeted promotional strategies.

Query 6 — Average Days Between Orders Per User

Objective: Calculate avg days between consecutive orders per customer | Customer Behavior

PYTHON CODE

```
q6_avg_days = (  
    orders[orders["days_since_prior_order"].notna()]  
    .copy()  
    .assign(days_since_prior_order=lambda df:  
        pd.to_numeric(df["days_since_prior_order"]))  
    .groupby("user_id", as_index=False)  
    .agg(avg_days_between_orders=("days_since_prior_order", "mean"))
```

User ID	Avg Days Between Orders
1	19.00
2	16.29
3	12.00
4	17.00
5	11.50
6	13.33
7	10.45
8	23.33

Interpretation

Average days between orders varies across customers. Some order every **10-12 days** (frequent shoppers), while others wait **20+ days** (occasional shoppers). This variation suggests distinct customer segments with different purchasing habits and engagement levels.

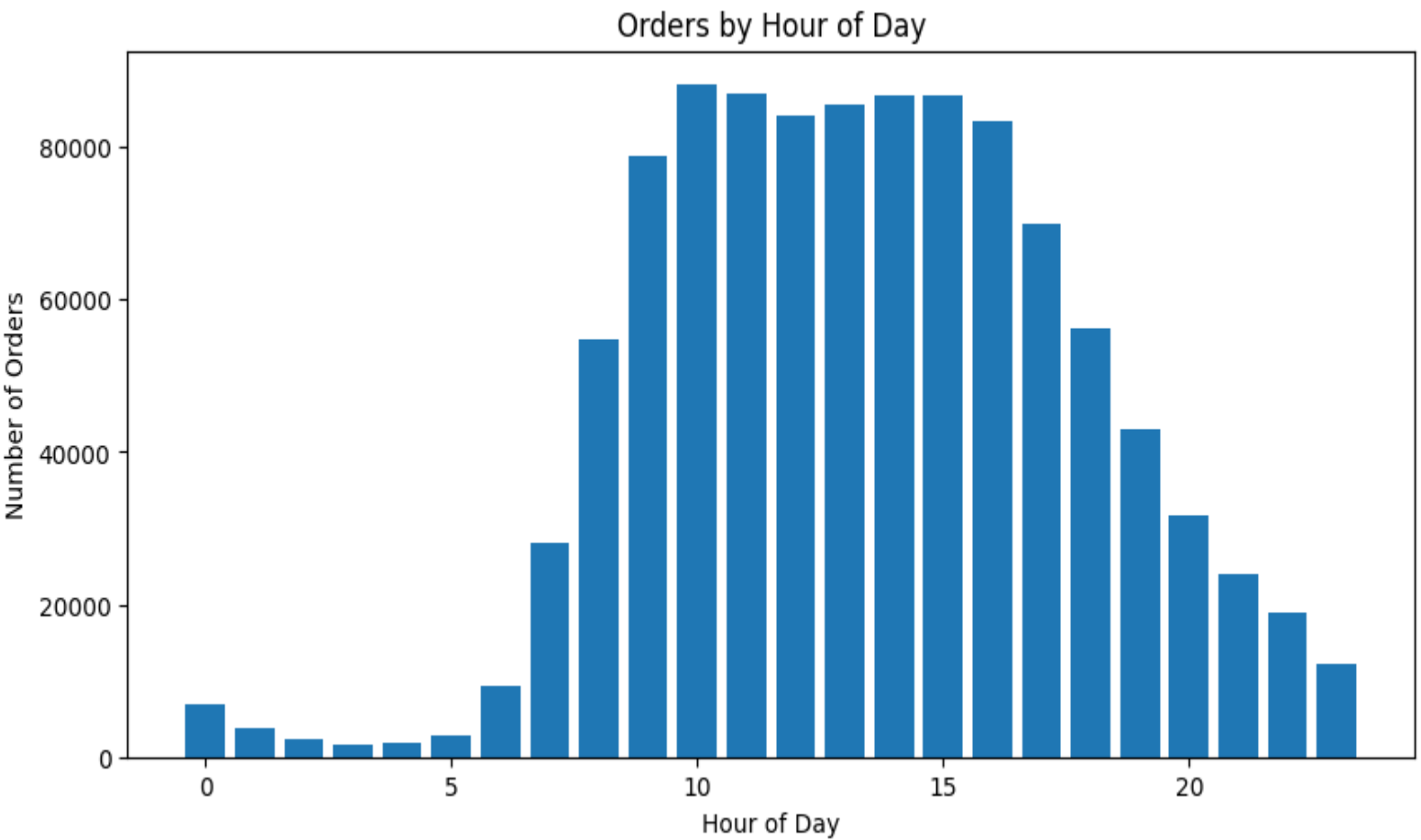
Business Insight: Customers with shorter ordering intervals are more engaged and respond well to loyalty programs. Customers with longer intervals can be targeted through personalized reminders and promotional offers to encourage repeat purchases.

Query 7 — Peak Hours of Order Placement

Objective: Identify highest order volume hours | Temporal / Demand-Timing Objective

PYTHON CODE

```
q7 = orders.groupby("order_hour_of_day", as_index=False)
        .agg(total_orders=("order_id","count"))
        .sort_values("total_orders", ascending=False)
plt.bar(q7["order_hour_of_day"], q7["total_orders"])
```



Interpretation

Customer ordering is highest between **10:00 AM and 4:00 PM**. The single busiest hour is **10:00 AM** with **88,228 orders**. Order volume gradually decreases after 4:00 PM. Lowest activity occurs during early morning hours (12:00 AM to 5:00 AM), when customer activity is minimal.

Business Insight: Customer demand is concentrated during late morning and afternoon, indicating predictable shopping behaviour. This information can be used to optimize staffing levels, warehouse operations, inventory replenishment schedules, and delivery planning during peak demand periods.

Query 8 — Order Volume by Day of Week

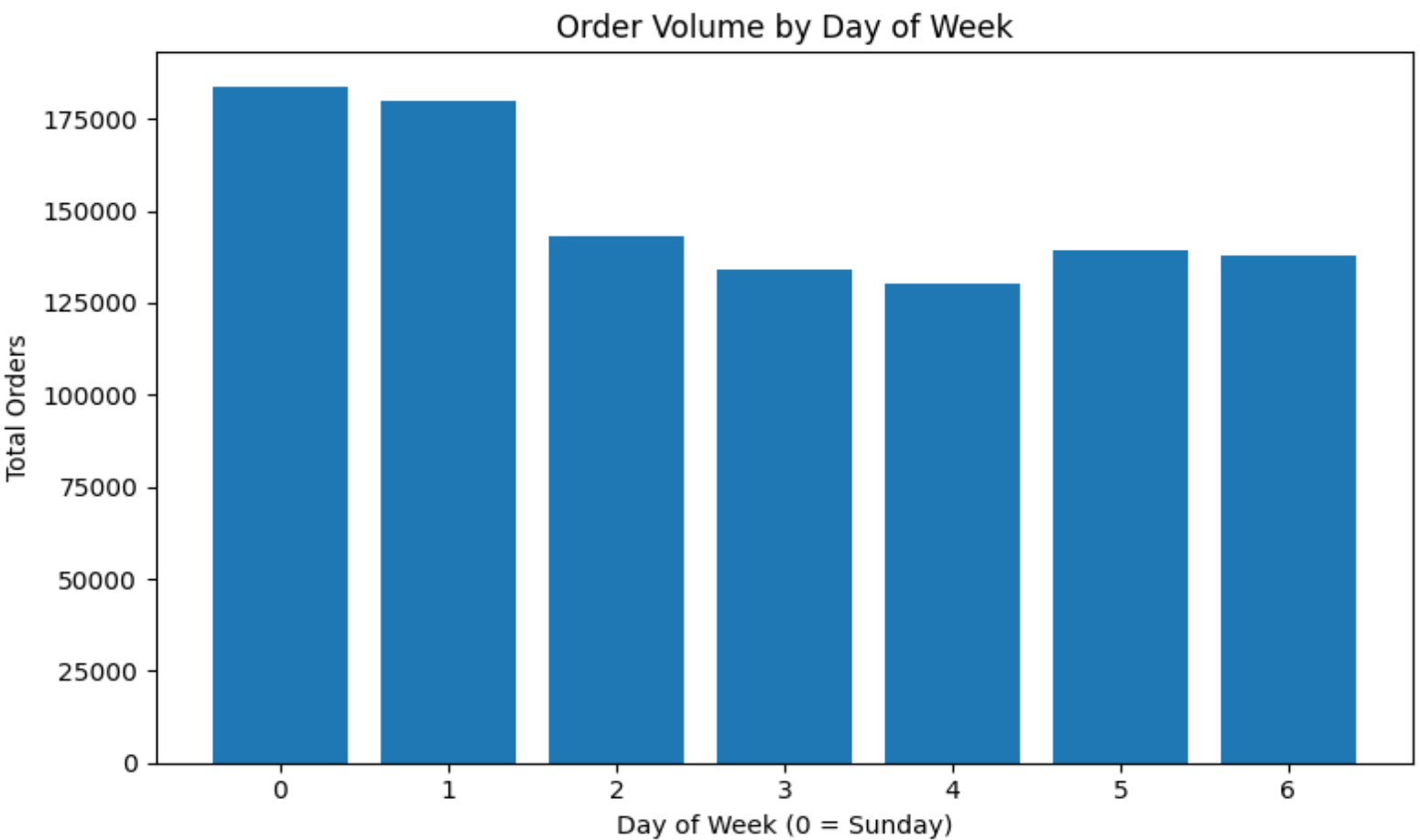
Objective: Analyze order distribution across days | Temporal / Demand-Timing Objective

PYTHON CODE

```
q8 = orders.groupby("order_dow", as_index=False)
      .agg(total_orders=("order_id","count"))
      .sort_values("order_dow")
plt.bar(q8["order_dow"], q8["total_orders"]); plt.xticks(range(7))
```

Interpretation

The highest number of orders was placed on **Sunday (183,939)**, followed closely by **Monday (180,025)**. Order volume gradually decreases from Tuesday to Thursday. A slight increase is observed on Friday and Saturday. Customer purchasing activity is concentrated at the beginning of the week.



Business Insight: Sunday and Monday are the busiest shopping days — critical periods for inventory availability and operational planning. Ensure adequate stock levels, workforce allocation, and delivery capacity during these peak days. Schedule promotional campaigns around high-traffic days.

Query 9 — Top 10 Most Ordered Products

Objective: Identify most frequently purchased products | Product Performance & Loyalty Objective

PYTHON CODE

```
q9_top_ordered = (  
    master_df.groupby("product_name", as_index=False)  
    .agg(total_orders=("product_id", "count"))  
    .sort_values("total_orders", ascending=False)  
    .head(10))
```

7 of top 10 are organic/fresh produce items

Interpretation

Banana is the most ordered product (4,357). Fresh fruits and vegetables dominate the list. Organic products (Organic Bananas, Strawberries, Spinach, Avocados) appear multiple times, indicating strong preference for organic items.

#	Product Name	Total Orders	Category
1	Banana	4,357	Fresh Fruit
2	Bag of Organic Bananas	3,522	Organic
3	Organic Strawberries	2,520	Organic
4	Organic Baby Spinach	2,286	Organic
5	Large Lemon	1,841	Fresh
6	Organic Hass Avocado	1,705	Organic
7	Organic Avocado	1,661	Organic
8	Strawberries	1,479	Fresh
9	Limes	1,383	Fresh
10	Organic Raspberries	1,279	Organic

Business Insight: Fresh produce and organic products are the primary demand drivers. Maintaining adequate inventory for these high-demand products is essential to prevent stock shortages. These products can be leveraged in promotional campaigns, bundle offers, and cross-selling strategies.

Query 10 — Unique Users by Department

Objective: Identify customer reach across departments | Customer Behavior Objective

PYTHON CODE

```
q10 = (master_df.groupby("department", as_index=False)
      .agg(unique_users=("user_id","nunique"))
      .sort_values("unique_users", ascending=False))
```

Top 3 departments reach 89.4% of all customers (produce + dairy eggs + beverages = 56,480 unique users)

Interpretation

Produce has the highest customer reach (22,303 users, 35.3%). Dairy Eggs follows (20,153). Essential grocery departments attract the largest customer bases. Bulk, Other, Pets, and Alcohol have comparatively fewer customers.

#	Department	Unique Users	Reach %
1	produce	22,303	35.3%
2	dairy eggs	20,153	31.9%
3	beverages	14,024	22.2%
4	snacks	13,304	21.1%
5	frozen	11,742	18.6%
6	pantry	10,886	17.3%
7	bakery	8,326	13.2%
8	deli	7,430	11.8%
...	...	...	...
18	alcohol	679	1.1%
19	pets	622	1.0%
20	other	392	0.6%
21	bulk	323	0.5%

Business Insight: Essential grocery departments (Produce, Dairy Eggs, Beverages, Snacks, Frozen) should receive higher priority for inventory planning, promotional campaigns, and shelf space. Low-engagement departments may require assortment optimization or targeted marketing.

Query 11 — Average Number of Products Per Order

Objective: Calculate average basket size | Customer Behavior Objective

PYTHON CODE

```
q11_avg_products = master_df.groupby("order_id").size().mean()
print(f"Average Products per Order: {q11_avg_products:.2f}")
```

Output: Average Products per Order: 10.51

10.51

Products Per Order (Average)

Interpretation

Customers purchase an average of **10.51 products per order**. This indicates most customers buy multiple products in a single session rather than making single-item purchases. The result suggests a **moderate basket size** reflecting planned grocery shopping behavior.

Basket Size Context:

- Small basket: 1-5 items (impulse/quick shop)
- Medium basket: 6-15 items (regular grocery)
- Large basket: 16+ items (stock-up shopping)

At 10.51 items, the average customer falls into the **medium basket** category, indicating regular planned grocery trips.

Business Insight: An average basket size of ~11 products presents opportunities for cross-selling and upselling. Retailers can increase average order value by recommending complementary products, offering bundle discounts, or promoting frequently-bought-together items during checkout.

Query 12 — Most Reordered Product by Department

Objective: Identify highest repeat-purchase product per department | Product Performance & Loyalty

PYTHON CODE

```
q12 = (master_df.groupby(["department",
    "product_name"], as_index=False)
    .agg(reorder_count=("reordered","sum")))
q12 = q12.sort_values(["department",
    "reorder_count"], ascending=[True,False])
q12.drop_duplicates("department")
```

Every department has a "hero product" driving loyalty

Interpretation

Banana dominates with 3,840 reorders in Produce. Organic Whole Milk (957) and Sparkling Water (637) show strong loyalty. Each department has a distinct flagship product. Pets, Alcohol, and Babies have lower reorder counts.

Department	Most Reordered Product	Reorder Count
produce	Banana	3,840
dairy eggs	Organic Whole Milk	957
beverages	Sparkling Water Grapefruit	637
deli	Original Hummus	539
bakery	100% Whole Wheat Bread	412
frozen	Blueberries	354
pantry	Extra Virgin Olive Oil	257
canned goods	Organic Black Beans	215
meat seafood	Ground Turkey Breast	170
...	...	...

Business Insight: Flagship products of each department should be priority items for inventory management, promotional campaigns, and shelf placement. These products consistently drive repeat purchases and customer retention across all categories.

Query 13 — Products Reordered More Than Once

Objective: Measure repeat purchase product count | Product Performance & Loyalty Objective

PYTHON CODE

```
q13 = master_df.groupby("product_name")
        .agg(reorder_count=("reordered","sum"))
q13_products = (q13["reorder_count"] > 1).sum()
print(f"Products Reordered More Than Once: {q13_products}")
```

Output: 11529

Interpretation

A total of **11,529 products** were reordered more than once. This indicates a significant portion of the catalogue experiences repeat purchases. Customers exhibit consistent purchasing behavior across many products, demonstrating product loyalty.



Out of **49,355 total products**, 11,529 products (approximately **23.4%** of the catalogue) have been reordered more than once. This indicates that nearly a quarter of the product range generates repeat purchase behavior, while the remaining ~76% are either one-time purchases or have very low reorder frequency.

Key Metrics

- 11,529 / 49,355 products reordered = **23.4%**
- ~76.6% of products have zero or one reorder
- Loyalty concentrated in perishable staples

Business Insight: A large number of repeatedly purchased products indicates strong customer loyalty. The business should prioritize maintaining inventory for these products, use them in personalized recommendations, loyalty programs, and targeted promotional campaigns to maximize repeat sales.

Query 14 — Average Cart Size Per Order

Objective: Determine average products added to cart | Customer Behavior Objective

PYTHON CODE

```
q14_avg_cart = master_df.groupby("order_id")
    .agg(cart_size=("add_to_cart_order","max"))
print(f"Average Cart Size: {q14_avg_cart['cart_size'].mean():.2f}")
```

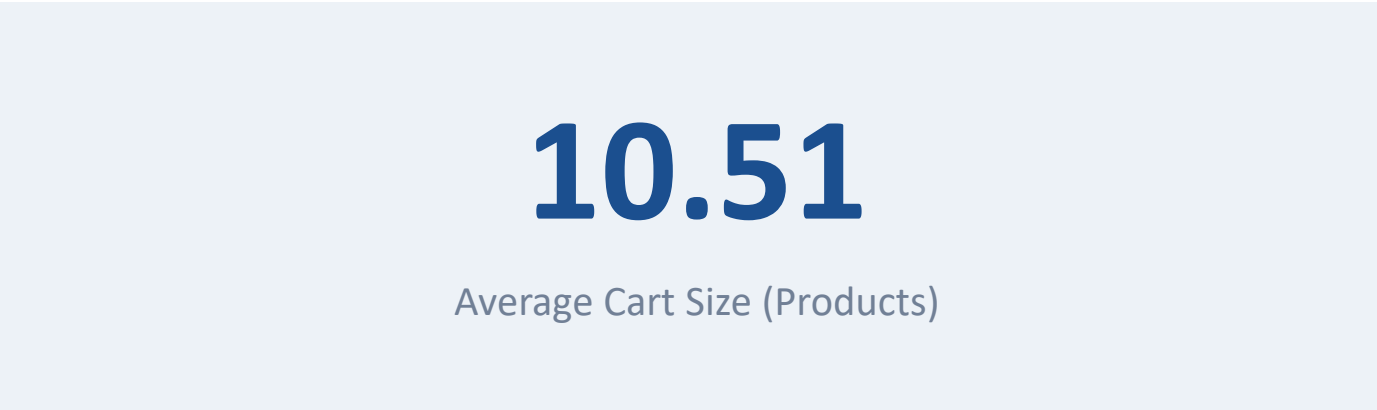
Output: Average Cart Size: 10.51

Note: Since add_to_cart_order reaches the maximum item position in an order, **both Query 11 and Query 14 produce the same result (10.51).** This cross-validation confirms the consistency of the analytical approach.

Interpretation

On average, customers add approximately **10.51 products** to their cart per order. The result reflects a moderate basket size suggesting planned grocery shopping behavior. The consistency with Query 11 validates the data integrity.

Business Insight: The average cart size indicates opportunities for increasing basket value through cross-selling and product recommendations. Bundle offers, complementary product suggestions, or volume-based discounts could encourage customers to add more products per order.



Query 15 — Orders by Hour of Day (Cross-Reference)

Objective: Identify order distribution across hours | Temporal / Demand-Timing Objective

Reference: Query 7 Results

Cross-Reference Note: This query requires the same analysis performed in **Query 7 (Peak Hours of Order Placement During the Day)**. Since both questions analyse the distribution of orders across the 24-hour period, the same Python implementation and results have been reused to avoid duplication.

The distribution indicates customers primarily place orders during regular daytime business hours, suggesting planned grocery shopping rather than impulse purchases.

Hour	Total Orders	Rank
10 AM	88,228	#1
11 AM	87,087	#2
2 PM	86,905	#3
3 PM	86,888	#4
1 PM	85,652	#5
12 PM	84,204	#6
4 PM	83,421	#7

Interpretation

Customer ordering activity is highest between **10:00 AM and 4:00 PM**. The single busiest hour is **10:00 AM** with **88,228 orders**. Order volume gradually decreases after 4:00 PM. The lowest ordering activity occurs during early morning hours (12:00 AM to 5:00 AM).

Business Insight: Peak hours (10 AM-4 PM) align with medium basket sizes (~10 products), confirming planned grocery shopping behaviour. Schedule staff during peak hours, launch promotional notifications before peak windows, and allocate delivery resources accordingly.

Query 16 — Distribution of Order Sizes

Objective: Analyze frequency of small, medium, and large orders | Temporal / Demand-Timing

PYTHON CODE

```
q16 = (master_df.groupby("order_id")
      .agg(order_size=("product_id","count"))
      .groupby("order_size", as_index=False)
      .agg(number_of_orders=("order_size","count"))
      .sort_values("order_size"))
```

Order Size	# of Orders	Category
1	1,536	Small
2	1,742	Small
3	1,883	Small
4	1,890	Small
5	2,032	Medium
...	...	...
66	1	Very Large
74 (max)	1	Very Large

Interpretation

Smaller and medium-sized orders occur much more frequently than very large orders. As order size increases, the number of orders gradually decreases. Orders containing more than 60 products are extremely rare. Most customers prefer purchasing a moderate number of products per order.

Key Statistics:

- Order sizes range from **1 to 74 products**
- **67 distinct order sizes** observed
- Most frequent: 5-product orders (2,032 orders)
- Orders 60+ products extremely rare (1 each)

Business Insight: Most customers prefer small-to-medium baskets, while exceptionally large orders are uncommon. Optimize inventory planning and fulfillment for typical basket sizes. Bundle offers and cross-selling may encourage larger baskets and increase average order value.

Query 17 — Average Reorder Rate by Aisle

Objective: Identify aisles with highest repeat purchase behavior | Product Performance & Loyalty

PYTHON CODE

```
q17 = master_df.groupby("aisle", as_index=False)
    .agg(avg_reorder_rate=("reordered","mean"))
q17["avg_reorder_rate"] = (q17["avg_reorder_rate"]
    * 100).round(2)
q17 = q17.sort_values("avg_reorder_rate", ascending=False)
```

Interpretation

The **Milk** aisle has the highest reorder rate (79.30%). Essential categories (Water, Fresh Fruits, Eggs) show 70%+ rates. In contrast, discretionary categories (Kitchen Supplies 16.33%, Spices 17.65%, Beauty 20%) have much lower rates. **Essential grocery categories achieve 3-4x higher reorder rates** than discretionary ones.

Top 5 Aisles

#	Aisle	Reorder Rate
1	milk	79.30%
2	water seltzer sparkling	73.97%
3	fresh fruits	73.50%
4	eggs	73.46%
5	packaged produce	70.98%

Bottom 5 Aisles

#	Aisle	Reorder Rate
130	kitchen supplies	16.33%
129	spices seasonings	17.65%
128	baking supplies decor	18.25%
127	beauty	20.00%
126	cold flu allergy	21.11%

Business Insight: Essential grocery categories consistently achieve higher customer retention and repeat purchases. Prioritize inventory planning and promotional campaigns for high-loyalty aisles. Explore initiatives to improve engagement in lower-performing discretionary categories.

Query 18 — Average Order Size by Day of Week

Objective: Analyze basket size variation across days | Temporal / Demand-Timing Objective

PYTHON CODE

```
q18 = (master_df.groupby(["order_dow","order_id"])
      .agg(order_size=("product_id","count"))
      .reset_index()
      .groupby("order_dow", as_index=False)
      .agg(avg_order_size=("order_size","mean"))
      .sort_values("order_dow"))
```

Range: 9.67 to 11.72 products (21% variation between min and max)

Day (0=Sunday)	Avg Order Size	Basket Type
Day 0 (Sunday)	11.72	Large
Day 6 (Saturday)	11.03	Large
Day 1 (Monday)	10.29	Medium
Day 5 (Friday)	10.20	Medium
Day 2 (Tuesday)	9.91	Medium
Day 4 (Thursday)	9.85	Medium
Day 3 (Wednesday)	9.67	Medium

Interpretation

Largest orders on **Day 0 (11.72)** and **Day 6 (11.03)** — weekend stock-up shopping. Smallest on **Day 3 (9.67)**. Variation is moderate (9.67-11.72), indicating relatively consistent purchasing behaviour with slight weekend uplift.

Business Insight: Customers tend to purchase larger baskets on weekends (Day 0 and Day 6), suggesting major grocery shopping days. Ensure adequate inventory, increase staff availability, and promote bundled offers during high-volume shopping days to maximize sales.

Query 19 — Top 10 Users with Highest Number of Orders

Objective: Identify most active customers | Customer Behavior Objective

PYTHON CODE

```
q19_top_users = (orders.groupby("user_id", as_index=False)
    .agg(total_orders=("order_id","count"))
    .sort_values("total_orders", ascending=False)
    .head(10))
```

Note: 100 orders/user may represent the dataset cap, not actual maximum purchasing.

Interpretation

Top 10 customers each placed exactly **100 orders**. The identical order count suggests **100 may be the dataset maximum** rather than a natural pattern. These customers demonstrate consistent, highly engaged purchasing behaviour.

Rank	User ID	Total Orders
1	15,896	100
2	16,290	100
3	57,838	100
4	57,851	100
5	16,298	100
6	16,313	100
7	16,340	100
8	16,357	100
9	16,364	100
10	964	100

Business Insight: Highly active customers represent the most valuable segment. They are ideal candidates for loyalty programs, personalized offers, premium memberships, and targeted retention campaigns. Understanding their behavior can improve customer lifetime value.

Query 20 — Products in Each Aisle and Department

Objective: Determine product distribution across aisles within departments | Catalogue & Assortment

PYTHON CODE

```
q20 = (products.merge(aisles, on="aisle_id")
        .merge(departments, on="department_id")
        .groupby(["department","aisle"], as_index=False)
        .agg(total_products=("product_id","count"))
        .sort_values(["department","total_products"],
                      ascending=[True,False]))
```

Interpretation

Certain aisles contain significantly more products than others within the same department. Alcohol: Beers Coolers (384) vs Specialty Wines (95). Snacks: Fruit Veg Snacks (356) vs Trail Mix (69). Product availability is concentrated in a few major aisles per department.

Total of **134 department-aisle combinations** identified across 21 departments.

Department	Aisle	Total Products
alcohol	beers coolers	384
alcohol	red wines	232
alcohol	spirits	192
snacks	fruit vegetable snacks	356
snacks	popcorn jerky	315
snacks	mint gum	168
snacks	ice cream toppings	83
snacks	trail mix snack mix	69

Business Insight: Understanding product distribution at department and aisle levels enables better category management, shelf space optimization, and inventory planning. Large-assortment aisles need greater merchandising attention; low-volume aisles can be evaluated for optimization.

09

Power BI Dashboards

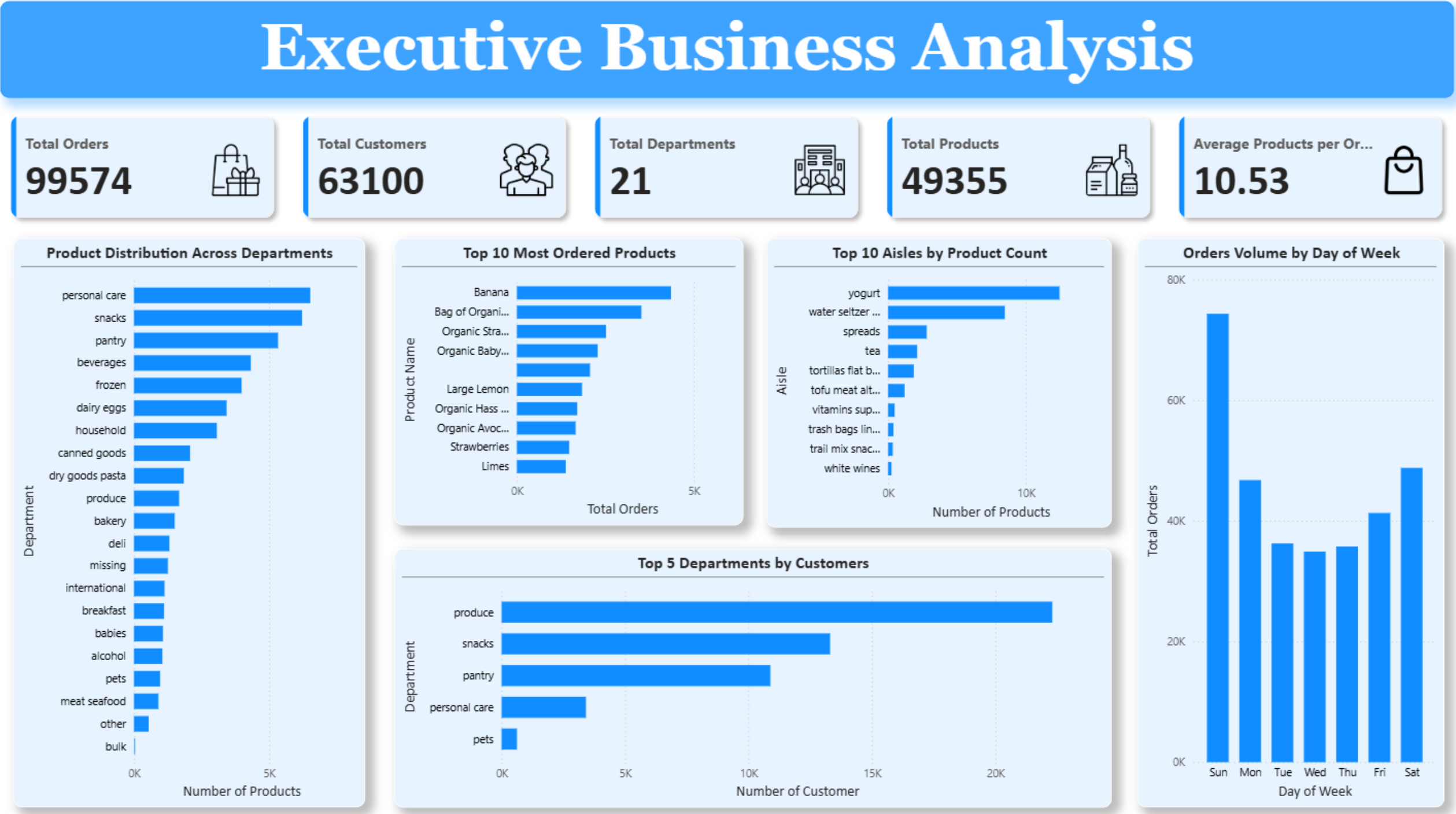
Four Interactive Executive Dashboards for Stakeholder Self-Service

After completing SQL and Python analysis, **four interactive dashboards** were developed in Power BI to present findings in a clear, business-friendly format for non-technical stakeholders.

Power BI — Executive Business Analysis Dashboard

Provides a **high-level summary** of the supermarket's overall performance. Combines key business metrics with product, customer, and order insights for quick management understanding.

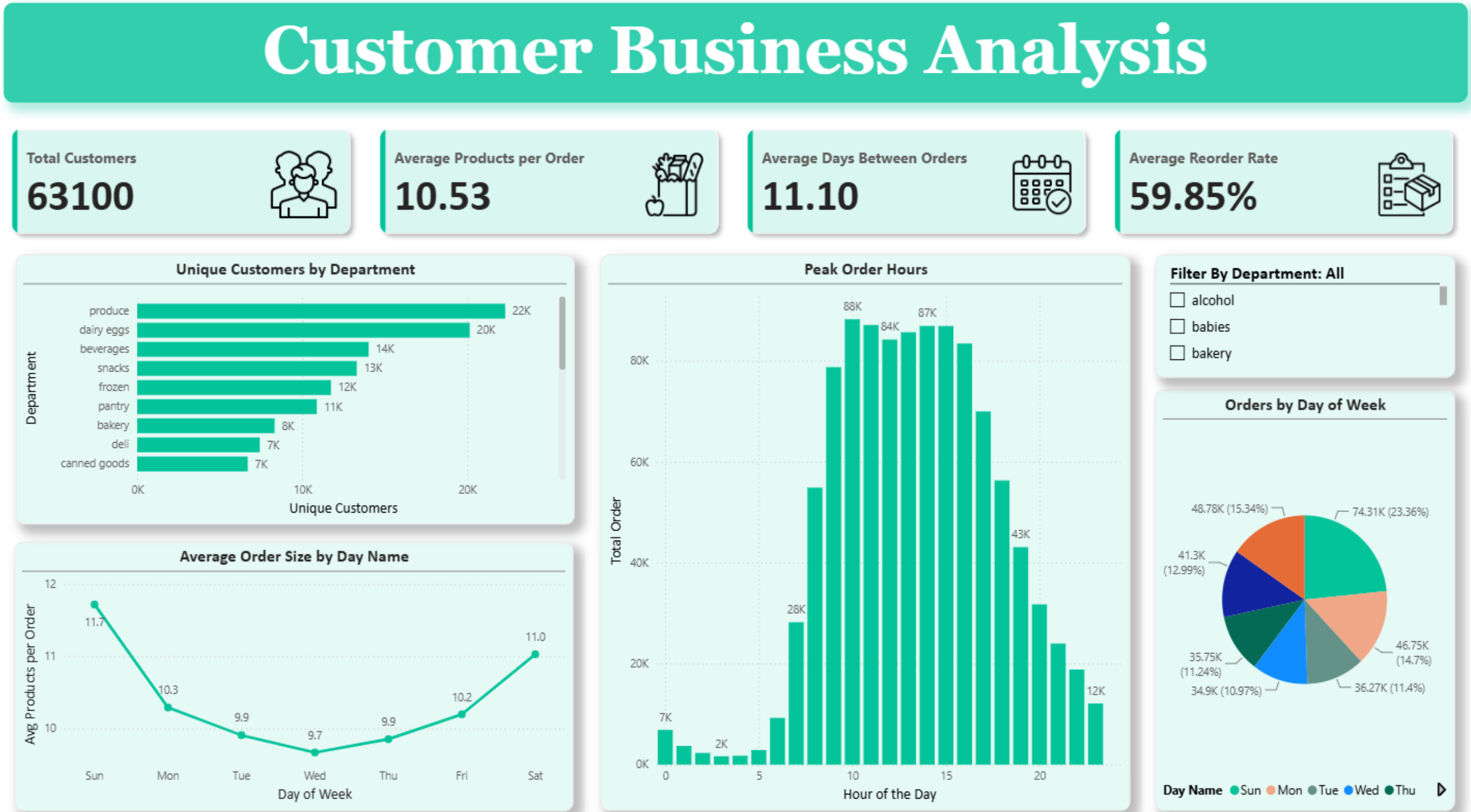
EXECUTIVE BUSINESS ANALYSIS DASHBOARD



Power BI — Customer Business Analysis Dashboard

Focuses on **customer purchasing behavior and shopping patterns**. Provides insights into customer distribution, ordering habits, reorder behavior, and shopping trends across departments.

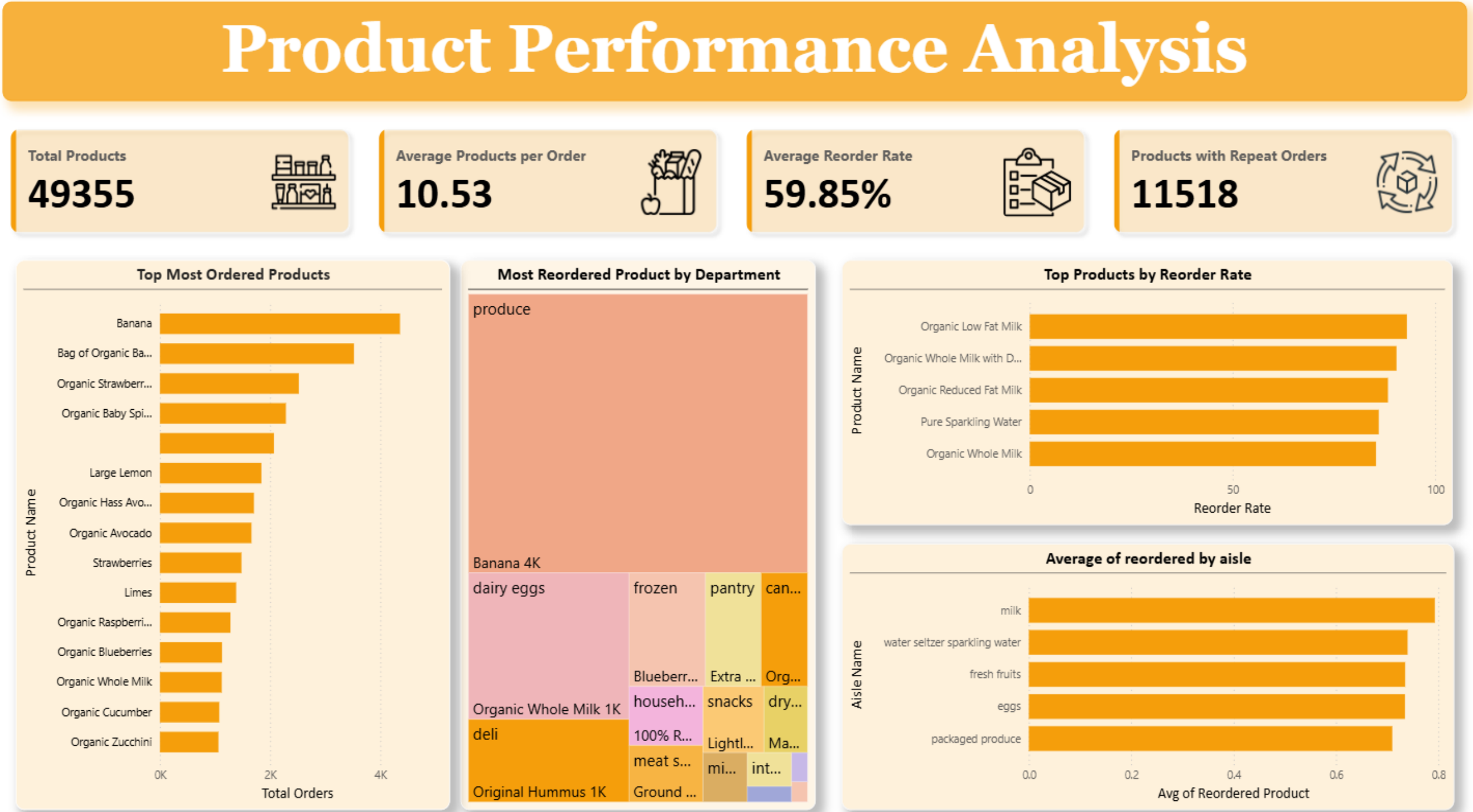
CUSTOMER BUSINESS ANALYSIS DASHBOARD



Power BI — Product Performance Analysis Dashboard

Evaluates **product-level performance** by analyzing purchasing patterns, reorder behavior, and demand across departments. Highlights best-performing products and identify customer loyalty.

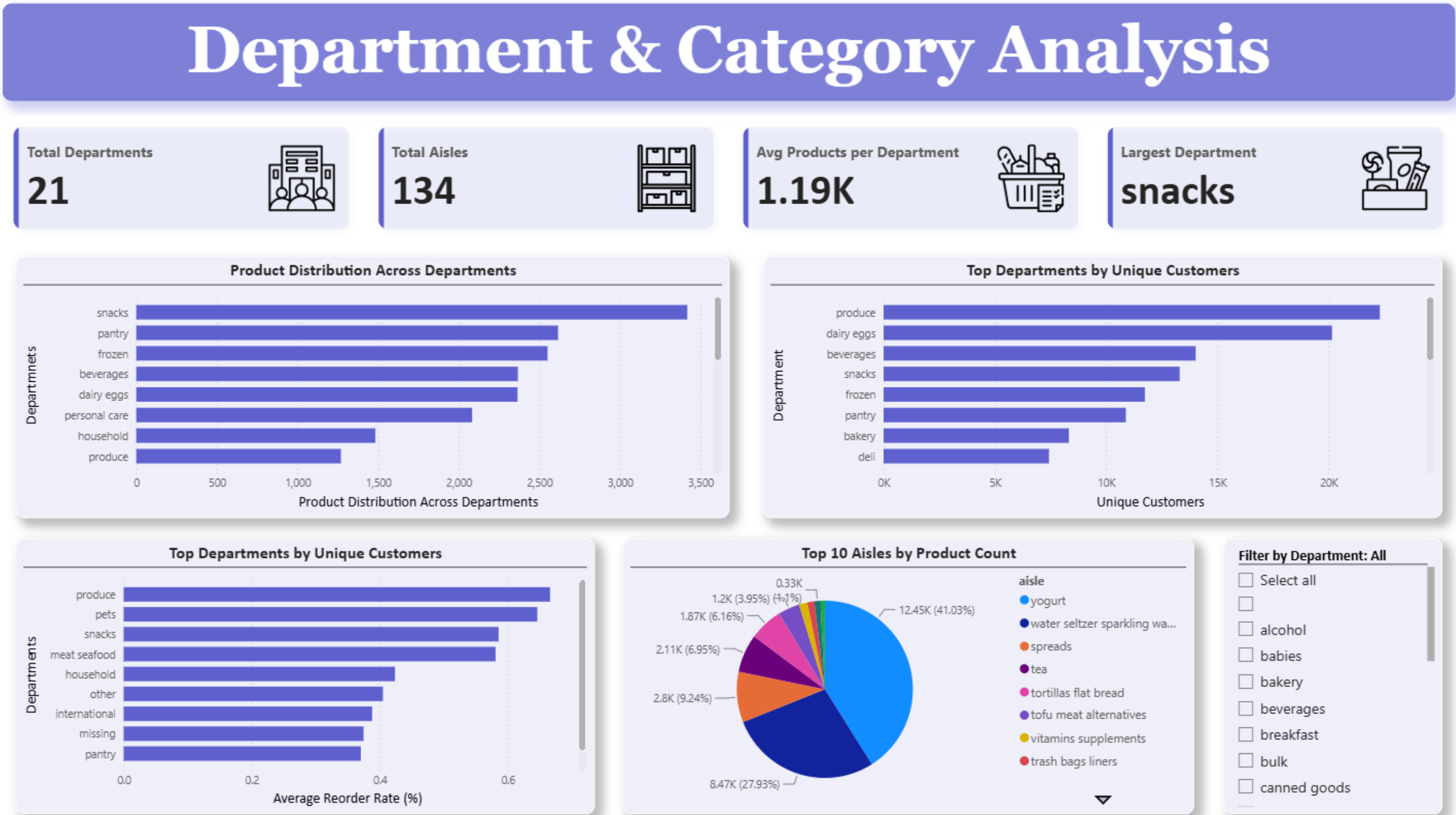
PRODUCT PERFORMANCE ANALYSIS DASHBOARD



Power BI — Department & Category Analysis Dashboard

Provides a **detailed view of product distribution and customer engagement** across departments and aisles. Helps evaluate category performance and identify high-performing departments

DEPARTMENT & CATEGORY ANALYSIS DASHBOARD



10

Key Findings & Business Insights

Cross-Analysis Variables, Product Portfolio, Customer Base,
Purchasing Behaviour, Product Loyalty, Demand & Reordering
Patterns

Cross-referencing **SQL validation**, **Python behavioral analysis**, and **Power BI dashboards** produced a consistent, triangulated view of the customer base and product performance.

Key Variables from Cross Analysis

Six key cross-analysis findings emerged from triangulating SQL validation, Python analysis, and Power BI dashboard data:

1 Assortment Loyalty: Departments with the largest product assortment are **not necessarily** those with highest reorder rates. A wider assortment does not automatically create stronger customer loyalty.

2 Produce Punch: Produce has fewer products than Personal Care, yet serves the **largest number of unique customers** (22,303). Produce items are everyday essentials with broad appeal.

3 Organic Loyalty: Organic dairy products consistently appear among the highest reorder-rate products. Customers who buy organic dairy are **highly loyal** with repeat purchase rates exceeding 85%.

4 Planned Shopping: Peak hours (10 AM-3 PM) align with medium basket sizes (~10 products), indicating customers are doing **planned grocery shopping** rather than impulse purchases.

5 Fresh Produce Dominance: Fresh produce dominates both **most-ordered** and **most-reordered** products, reinforcing its importance in overall store performance and customer retention.

6 Strategic Focus: High-performing departments (Produce, Dairy Eggs, Beverages, Snacks) combine **large customer reach** with **strong repeat purchasing**, making them strategic focus areas.

Convergent Evidence: The single most decision-ready finding is the convergent evidence across four independent metrics (order volume, reorder rate, customer reach, basket composition) that a **small set of perishable staples anchors customer engagement**, while the majority of the 49,355-product catalogue sees minimal repeat engagement. This insight should drive the next round of assortment and retention-marketing planning.

Business Insights — Product Portfolio & Customer Base

49,355	21	63,100	10-25
Products in Catalogue	Departments	Unique Customers	Days Between Orders

A. Product Portfolio Insights

Product Catalogue and Distribution: 49,355 products across 21 departments and 134 aisles. Distribution is highly uneven — Personal Care (6,524), Snacks (6,221), Pantry (5,331) hold largest shares. Bulk is smallest with only 38 products.

Impact: Inventory management should focus on departments with large assortments. Smaller departments may require assortment optimization.

Aisle Variety: Certain aisles show extremely large variety — Missing (1,254), Candy Chocolate (1,235), Ice Cream (1,084), Vitamins & Supplements (1,030).

Impact: Large aisles require better shelf planning. Higher assortment increases customer choice but also operational complexity.

B. Customer Base Insights

Customer Base: The store maintains a large active customer base of **63,100 unique customers** in the dataset.

Impact: Large enough customer base for meaningful segmentation and personalized marketing campaigns.

Purchase Frequency: Customers purchase regularly, with average time between transactions typically ranging from **10 to 25 days**.

Impact: Recurring purchasing behaviour creates opportunity for reminder campaigns and subscription models.

Business Insights — Purchasing Behaviour & Product Loyalty

C. Purchasing Behaviour Insights

Midday Shopping Peak: Orders peak at 10 AM, 11 AM, 2 PM, 3 PM. Lowest volumes occur 2 AM-5 AM.

Impact: Schedule staff during peak hours. Launch promotional notifications before peak windows.

Basket Size: Customers purchase avg **10.51 products per order**. Order sizes range 1-74, but most are small-to-medium baskets.

Impact: Cross-selling opportunities exist. Bundle offers may encourage larger baskets.

Weekend Busier: Order volume peaks on Saturday and Sunday. Wednesday has lowest traffic.

Impact: Increase inventory and staffing during busy days.

D. Product Loyalty Insights

Organic Loyalty: Organic dairy products show exceptionally strong loyalty. Organic Low-fat Milk, Whole Milk, DHA, Reduced Fat Milk all maintain **reorder rates exceeding 85%** at 100-order threshold.

Impact: Organic products drive repeat purchases — expand and promote.

Department Hero Products: Each department has its own loyalty driver — Banana (Produce), Organic Whole Milk (Dairy), Original Hummus (Deli), Sauvignon Blanc (Alcohol).

Impact: Department-specific promotions should focus on these hero products as traffic drivers.

Bananas Dominate: Bananas rank as both highest-ordered and highest-reordered item in Produce department.

Impact: Core demand driver — inventory shortages directly affect satisfaction.

Business Insights — Customer Reach, Demand & Reordering

E. Customer Reach Insights

Produce Reaches the Largest Number of Customers: The Produce department reaches the largest customer base (22,303), followed by Dairy Eggs (20,153), Beverages (14,024), and Snacks (13,304) as categories with highest unique customer counts.

Impact: These departments attract the broadest customer base and deserve greater promotional investment.

F. Product Demand Insights

Fresh Produce Dominates Orders: Fresh produce dominates customer orders, with top-ordered products consisting of bananas, organic bananas, organic strawberries, organic baby spinach, avocados, and lemons.

Impact: Fresh produce drives customer traffic. Demand forecasting is especially important for perishables to prevent stockouts and waste.

G. Reordering Behaviour Insights

High Reorder Rates Need Context: While several products initially showed 100% reorder rate, these figures were skewed by low purchase volumes. Applying a minimum threshold of 100 orders established a much more reliable metric.

Impact: Business decisions should not rely solely on percentages; sufficient sample size is essential. Combine reorder rate with total order volume for reliable conclusions.

Insight Summary Matrix

Produce = highest reach (22,303 customers) + dominates orders + dominates reorders | **Organic dairy** = highest loyalty (85%+ reorder) | **Peak hours** = 10 AM-3 PM | **Peak days** = Sun-Mon | **Avg basket** = 10.51 items | **11,529 products** reordered multiple times

11

Recommendations & Impact

Twelve Data-Backed Recommendations for Marketing,
Merchandising, and Operations

Each recommendation is grounded in specific analytical findings and includes **expected business benefits** to enable prioritization by Marketing, Merchandising, and Operations teams.

Recommendations

A. Prioritize Inventory for High-Demand Products

Observation: Banana, Organic Bananas, Strawberries, Spinach, Milk top order/reorder lists.

Action: Maintain higher safety stock. Use demand forecasting. Monitor daily.

Benefit: Higher availability, increased satisfaction, reduced lost sales.

B. Strengthen Organic Product Portfolio

Observation: Organic dairy and produce have exceptionally high reorder rates (85%+).

Action: Expand organic offerings. Promote organic bundles. Feature in campaigns.

Benefit: Improved retention, higher repeat purchases, increased revenue.

C. Optimize Staffing During Peak Hours

Observation: Most orders occur between 10 AM and 3 PM (peak 88,228 at 10 AM).

Action: Increase staffing during peak. Schedule replenishment before peak.

Benefit: Faster service, reduced waiting time, better efficiency.

D. Increase Weekend Readiness

Observation: Order volume highest on Day 0 and Day 1 (Sun-Mon: 363,964 orders).

Action: Increase inventory before weekends. Schedule additional staff.

Benefit: Better customer experience, increased sales, reduced shortages.

E. Focus Promotions on High-Customer Departments

Observation: Produce, Dairy, Beverages, Snacks attract highest unique customers.

Action: Launch targeted promotions. Place featured products at entrances.

Benefit: Increased engagement, higher basket value, more cross-selling.

F. Improve Low-Performing Departments

Observation: Bulk, Pets, Alcohol, Other have fewer products and fewer customers.

Action: Review product assortment. Remove underperformers. Test new products.

Benefit: Better inventory utilization, lower holding costs, improved performance.

Recommendations

G. Increase Average Basket Size

Observation: Customers purchase avg 10.5 products per order.

Action: "Frequently Bought Together" recommendations. "Buy More Save More" promotions. Meal kits and bundles.

Benefit: Higher average order value, increased revenue per customer.

H. Build Customer Loyalty Programs

Observation: Many products have high reorder rates indicating strong repeat behaviour.

Action: Personalized loyalty rewards. Repeat-purchase discounts. Reorder reminders.

Benefit: Higher retention, increased lifetime value, more recurring sales.

I. Improve Shelf Space Allocation

Observation: Personal Care, Snacks, Pantry contain the largest number of products.

Action: Allocate shelf space based on product demand, not just product count. Regularly review category performance.

Benefit: Better visibility, improved shopping experience, increased sales.

J. Use Purchase Patterns for Marketing

Observation: Customers exhibit consistent purchasing cycles and predictable behaviour.

Action: Schedule campaigns before peak periods. Personalize offers. Recommend complementary products.

Benefit: Higher conversion rates, increased repeat purchases, better engagement.

K. Develop Department-Specific Marketing

Observation: Each department has a distinct top-performing product (e.g., Banana, Organic Milk).

Action: Design promotions around each department's hero product. Use as traffic drivers.

Benefit: Increased department-level sales, higher cross-selling opportunities.

L. Base Decisions on Reliable Metrics

Observation: Products with very few purchases can show misleadingly high reorder rates.

Action: Apply minimum order threshold (e.g., 100) before evaluating loyalty. Combine rate with volume.

Benefit: More reliable decision-making, reduced risk from weak patterns.

Expected Business Impact Summary

The 12 recommendations are expected to deliver measurable impact across six key business dimensions:



Operational Efficiency

- Faster service during peak hours
- Reduced customer waiting time
- Better resource allocation
- Optimized staffing schedules



Revenue Growth

- Higher average order value
- Increased sales from promotions
- More cross-selling opportunities
- Better promotional ROI



Customer Satisfaction

- Higher product availability
- Better shopping experience
- Reduced stockouts
- Faster checkout process



Customer Retention

- Higher retention rates
- Increased lifetime value
- More recurring sales
- Stronger brand loyalty



Inventory Optimization

- Better inventory utilization
- Lower holding costs
- Improved demand forecasting
- Reduced waste



Decision Quality

- More reliable metrics
- Reduced risk from weak patterns
- Data-driven strategy
- Evidence-based prioritization

Implementation Priority: Recommendations A (Inventory), C (Staffing), and D (Weekend Readiness) should be implemented first as they address immediate operational needs with clear, measurable outcomes. Recommendations G (Basket Size) and H (Loyalty Programs) follow as medium-term strategic initiatives.

Conclusion & Future Scope

Conclusion

This analysis converts a relational grocery-order dataset into a **validated, cross-checked picture** of what's bought, how often, by whom, and when — anchored by SQL-level integrity checks, complete pandas answers to all 20 brief questions, and four Power BI dashboards for stakeholder self-service.

The single most decision-ready finding is the **convergent evidence across four independent metrics** that a small set of perishable staples anchors customer engagement, while catalog breadth sits elsewhere.

Future Scope

- 1

Validate the extraction cap — confirm whether the ~1,048,575-row tables represent a capped extract and, if so, re-run this analysis against the full order history.
- 2

Enrich with pricing/margin data — add pricing, margin, and marketing-channel data so reorder-rate insights can be turned into ROI-ranked recommendations.
- 3

Extend to rolling/cohort view — analyze reorder cadence by customer tenure and segment to identify churn risk and loyalty progression patterns.
- 4

Resolve data quality items — address the "missing" category label and small metric discrepancies as a formal data-quality remediation backlog item.
- 5

Move to refreshable Power BI model — transition from a point-in-time CSV export to a refreshable Power BI model so the dashboards stay current as new orders arrive, rather than reflecting a single export snapshot.

Thank You

Market Analysis Project Report